SC-INR: An Internal Technical Record of a Constrained Cell Path in Fourier Implicit Arbitrary-Scale Image Super-Resolution

Frozen internal record, not for submission

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Route status (7 August 2026). The P1.12 source-and-fairness gate for both IPE-LIIF and Thera did not pass. P1.13 was not started, and the SC-INR new-method claim is closed. This document preserves implementation details and bounded empirical facts for internal reproducibility; it is not an active submission manuscript.

Abstract

Arbitrary-scale image super-resolution (ASISR) commonly represents an image as a local implicit function queried at continuous output coordinates. Prior work on integrated positional encoding (IPE) already models an output pixel as an area and analytically integrates fixed Fourier features over that area. Thera further combines content-conditioned sinusoidal fields with an analytic point spread function. Against this prior art, this draft studies a narrower question: whether an LTE-style content-adaptive Fourier decoder benefits from replacing its learned cell-conditioned phase with a constrained footprint response. The current SC-INR implementation gives small positive three-seed OOD mean deltas over LIIF and LTE under the current bicubic protocol. The pre-frozen provenance and fairness requirements for the two direct baselines were not met, so no direct-baseline training was started and the method-level claim was closed. The present document is therefore a frozen internal technical record, not a claim awaiting completion by adjacent experiments.

1 Introduction

ASISR aims to recover high-resolution images at arbitrary continuous scale factors. Local implicit image functions, represented by coordinate-conditioned MLPs over image features, provide a practical route to this setting [4]. LTE improves this formulation by predicting local Fourier coefficients and frequencies from image features, enabling stronger texture modeling than a plain MLP [6].

The current work revisits how the output cell enters such Fourier decoders. In the ASISR query protocol, the cell is not an arbitrary scale token: it specifies the output pixel footprint in the continuous image domain. In LTE, however, this cell is fed to a learned phase map and directly shifts the Fourier argument. This can improve fitting near training scales, but it also gives the network a shortcut for moving local texture phase as a function of the requested output footprint.

We therefore study decoder-side sampling consistency as a hypothesis. The central idea is to separate the feature-conditioned continuous content from the scale-conditioned observation. In SC-INR, the Fourier coefficient, frequency, and optional content phase are functions of local image features only. The output cell affects the decoded observation through an analytic sinc response derived from box averaging a Fourier component. This is not a claim of whole-network scale equivariance: the encoder, degradation model, and sampling protocol are not proven to commute with scale transformations. The implementation intervention recorded here is a constrained implicit decoder contract that removes arbitrary learned cell-conditioned phase extrapolation; the present evidence does not establish this intervention as a new method contribution.

2 Related Work and Prior-Art Boundary

IPE-LIIF treats each output pixel as an area rather than a center point and analytically integrates sine and cosine positional encodings over a rectangular pixel, yielding the same separable sinc form used by SC-INR [7]. Mip-NeRF introduced the broader idea of footprint-integrated positional encoding for anti-aliased neural fields [1]. Thera provides a stronger ASISR construction in which a content-conditioned local sinusoidal color field is observed through an analytic Gaussian point spread function [2]. Consequently, pixel-footprint modeling, analytic Fourier response, and arbitrary-scale generalization are prior art and are not claimed as SC-INR contributions.

Other ASISR work has pursued scale-aware feature extraction and upsampling [8], local attention and frequency encoding [3], and real-world continuous optical zoom data rather than pure bicubic degradation [5]. The remaining implementation distinction is LTE-specific: content-adaptive frequencies are retained while the explicit cell path is moved from learned phase to an analytic pre-MLP response. P1.12 did not pass the source-and-fairness requirements needed to evaluate this distinction against both direct baselines; it is therefore retained only as an internal implementation fact.

3 Method

3.1 Local Fourier Decoder

Let $E(I_{\mathrm{LR}})$ be the encoder feature map. For a query coordinate $\mathbf{x}$, the decoder samples a local feature $\mathbf{z}$ and forms a relative coordinate δ to the local feature anchor. LTE predicts coefficients and frequencies from $\mathbf{z}$, but also adds a learned phase term $h_p(\mathbf{c})$:

$$\cos\left(\pi(\boldsymbol{\omega}(\mathbf{z})^\top \delta + h_p(\mathbf{c}))\right). \tag{1}$$

This creates a cell-specific local function $f_{\mathbf{z},\mathbf{c}}$, rather than a single content function observed under different output footprints. Under scale extrapolation, this branch is unconstrained: a larger footprint can alter where the Fourier component appears, not merely how strongly it is observed.

3.2 Analytic Observation Response

Consider a one-dimensional Fourier component

$$f(x) = A\cos(\pi(\omega x + \phi)). \tag{2}$$

The box-averaged observation over a cell of width c is

$$\frac{1}{c}\int_{x_0-c/2}^{x_0+c/2} A\cos(\pi(\omega x+\phi))\,dx = A\cos(\pi(\omega x_0+\phi))\,\mathrm{sinc}\left(\frac{\omega c}{2}\right), \tag{3}$$

where $\mathrm{sinc}(t) = \sin(\pi t)/(\pi t)$. For a two-dimensional rectangular footprint, the separable response is

$$W(\boldsymbol{\omega}, \mathbf{c}) = \mathrm{sinc}\left(\frac{\omega_h c_h}{2}\right)\mathrm{sinc}\left(\frac{\omega_w c_w}{2}\right). \tag{4}$$

This is a signed sinc response. It can become negative for some frequency-footprint pairs, which corresponds to the signed response of a box filter rather than a guaranteed positive attenuation envelope.

3.3 SC-INR Variants

The frozen SC-INR implementation predicts content terms from features:

$$A = A(\mathbf{z}), \qquad \boldsymbol{\omega} = \boldsymbol{\omega}(\mathbf{z}), \qquad \phi = \phi(\mathbf{z}), \tag{5}$$

where A denotes the pre-MLP coefficient vector used by the decoder, not a direct measurement of the final RGB image spectrum. It uses signed bounded frequencies

$$\boldsymbol{\omega}(\mathbf{z}) = \omega_{\max}\tanh(g_\omega(\mathbf{z})). \tag{6}$$

The decoder input is

$$A(\mathbf{z}) \odot \left[\cos(\pi(\boldsymbol{\omega}(\mathbf{z})^\top \delta + \phi(\mathbf{z}))), \sin(\pi(\boldsymbol{\omega}(\mathbf{z})^\top \delta + \phi(\mathbf{z})))\right] \odot W(\boldsymbol{\omega}(\mathbf{z}), \mathbf{c}). \tag{7}$$

Equivalently, the coefficient part can be viewed as a feature-level effective amplitude

$$A_{\text{eff}}(\mathbf{z}, \mathbf{c}) = A(\mathbf{z}) \odot W(\boldsymbol{\omega}(\mathbf{z}), \mathbf{c}), \tag{8}$$

which separates content-dependent coefficient strength from the observation-dependent response of the output footprint. This interpretation applies to the Fourier feature vector passed to the MLP decoder; because SC-INR still uses an MLP, local ensemble, and residual upsampling, the full RGB predictor should not be described as an exact analytic box integral. The feature phase $\phi(\mathbf{z})$ does not receive cell or scale. The previous no-phase variant, denoted SC-INR-NoPhi, sets $\phi = 0$. We keep checkpoint and raw-result compatibility names separate from paper display names; in older result files, the raw key "SC-INR" refers to SC-INR-NoPhi, while the current frozen SC-INR implementation corresponds to raw key "SC-INR+PhiZ" in the seed1 historical file and to the cleaned raw key "SC-INR" in the seed2/seed3 files.

4 Experiments

4.1 Protocol

We use the inherited ASISR protocol with HR on-the-fly bicubic downsampling, avoiding the unreliable local pre-generated LR benchmark files. Models are trained on scales up to $4\times$ and evaluated at in-distribution scales $\{2, 3, 4\}$ and OOD scales $\{6, 8, 12, 16, 24, 30\}$. PSNR is reported on Set5, Set14, BSD100, and Urban100. Auxiliary diagnostics are currently computed on BSD100 and Urban100 subsets for seed1. Paper benchmark tables are generated from the canonical benchmark entry `artifacts/derived/benchmarks/`, which records the raw key and canonical model name for every row.

4.2 Seed1 Frozen-Implementation Benchmark

Table 1: Seed1 benchmark context. The frozen SC-INR implementation was strongest in seed1, but the current claim should use the three-seed summary in Table 3.

Model	ID PSNR	OOD PSNR	All PSNR
LIIF	30.9992	22.7111	25.4738
LTE	31.0450	22.6917	25.4761
SC-INR-NoPhi	31.0515	22.7471	25.5152
SC-INR-NoPhi-Signed	31.0567	22.7418	25.5134
SC-INR	31.1187	22.7766	25.5573
LIIF-EQ	31.1205	22.7469	25.5381

Table 1 records that the frozen SC-INR implementation has a positive seed1 OOD PSNR by +0.0849 dB over LTE and by +0.0295 dB over SC-INR-NoPhi. This is the intended place to show the SC-INR vs SC-INR-NoPhi structural increment.

4.3 Core Multi-Seed Result

Table 2: Core three-seed benchmark for LIIF, LTE, and SC-INR-NoPhi. These values record behavior under the current protocol and do not establish a general method advantage.

Model	Split	Mean PSNR	Std.	Delta vs LTE
LIIF	ID	31.0301	0.0274	-0.0485
LIIF	OOD	22.7251	0.0124	+0.0177
LTE	ID	31.0786	0.0296	+0.0000
LTE	OOD	22.7074	0.0137	+0.0000
SC-INR-NoPhi	ID	31.0746	0.0203	-0.0040
SC-INR-NoPhi	OOD	22.7581	0.0105	+0.0507

Table 2 records a small positive OOD mean delta for SC-INR-NoPhi over LTE, while its ID average is essentially tied with LTE. This is a bounded observation under the current protocol; it does not establish general scale-extrapolation improvement or sampling consistency.

4.4 Frozen SC-INR Multi-Seed Result

Table 3: Frozen SC-INR three-seed benchmark. Deltas are paired seed deltas against LIIF and LTE under the same benchmark protocol.

Split	Mean PSNR	Std.	Delta vs LIIF	Delta vs LTE
ID	31.0738	0.0582	+0.0437	-0.0048
OOD	22.7578	0.0316	+0.0328	+0.0504
All	25.5298	0.0403	+0.0364	+0.0320

Table 3 records positive OOD and All-scale mean deltas over LIIF and LTE under this three-seed protocol, while the ID delta versus LTE is slightly negative. The deltas are small relative to seed variation and do not establish a stable or general method advantage. SC-INR versus SC-INR-NoPhi remains seed1 context rather than a multi-seed attribution claim.

4.5 Empirical Cross-Grid Agreement Diagnostic

Table 4: Seed1 same-LR empirical cross-grid agreement. Larger PSNR indicates closer agreement after bicubic resizing larger-scale predictions to the $4\times$ grid under the same LR input. This is not a box-observation test.

Model	Dataset	Consistency PSNR	Delta vs LTE
LTE	BSD100	44.1914	+0.0000
SC-INR-NoPhi	BSD100	52.8053	+8.6140
LTE	Urban100	39.2983	+0.0000
SC-INR-NoPhi	Urban100	47.0694	+7.7712

The large values in Table 4 show that removing the learned cell-conditioned phase changes cross-grid behavior. They do not establish sampling correctness: the comparison uses bicubic resizing rather than the declared box observation operator, and cell-insensitive controls can also score highly. This diagnostic remains *diagnostic-only*.

4.6 Unavailable Historical Diagnostics

Earlier drafts included footprint-oracle and unified mechanism tables whose canonical artifacts are currently missing. Those tables have been removed from this frozen record. They cannot return through routine reruns; any future use requires a new Node Contract, restored canonical artifacts, artifact-whitelist registration, and controls that reject trivial cell-insensitive or incorrect-response behavior.

4.7 Selected Qualitative Examples

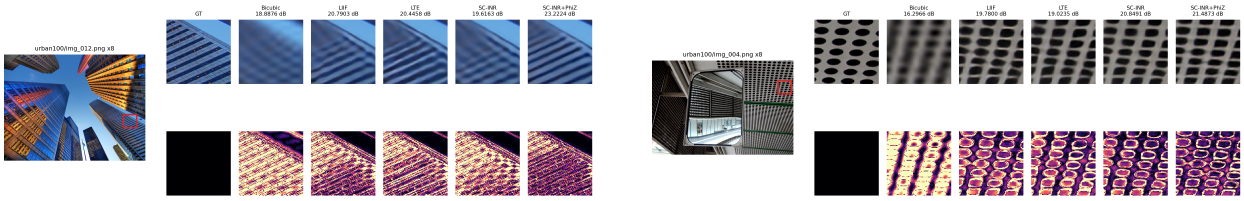


Figure 1: User-confirmed selected Urban100 8× examples where the frozen SC-INR implementation shows clearer local structure than LTE. These are selected examples, not evidence of average visual quality.

5 Limitations and Next Steps

The primary limitation is prior-art coverage. IPE-LIIF already establishes pixel-area integrated Fourier encoding in ASISR, while Thera provides a stronger analytic field-observation construction. SC-INR has not yet been compared to either under the same protocol. Its sinc response is applied before a nonlinear MLP, local ensemble, and residual path, so the final RGB predictor is not an exact box integral. The current cross-grid metric uses bicubic resizing and can reward cell-insensitive behavior. The synthetic bicubic benchmark also does not establish behavior under a measured real camera PSF; COZ provides one relevant real-world direction. The P1.12 audit found no verifiable official IPE-LIIF software provenance and no immutable Thera checkpoint provenance; Thera's model-native training recipe is also not a strict LTE-style factorial. Because the pre-frozen dual-baseline gate failed, P1.13 was not started and this route was closed rather than extended with adjacent experiments.

6 Conclusion

SC-INR is an LTE-specific constrained cell-path implementation, not an established new sampling principle. Its three-seed benchmark records small positive OOD mean deltas over LIIF and LTE under the current protocol, while the existing cross-grid metric and mechanism studies remain diagnostic. The frozen dual-baseline gate did not pass, no follow-on training was launched, and the SC-INR new-method claim is closed. The code and results are retained only as an internal reproducibility and mechanism case study. New external provenance could justify a new audit, but would not by itself restore novelty or a method claim.

References

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